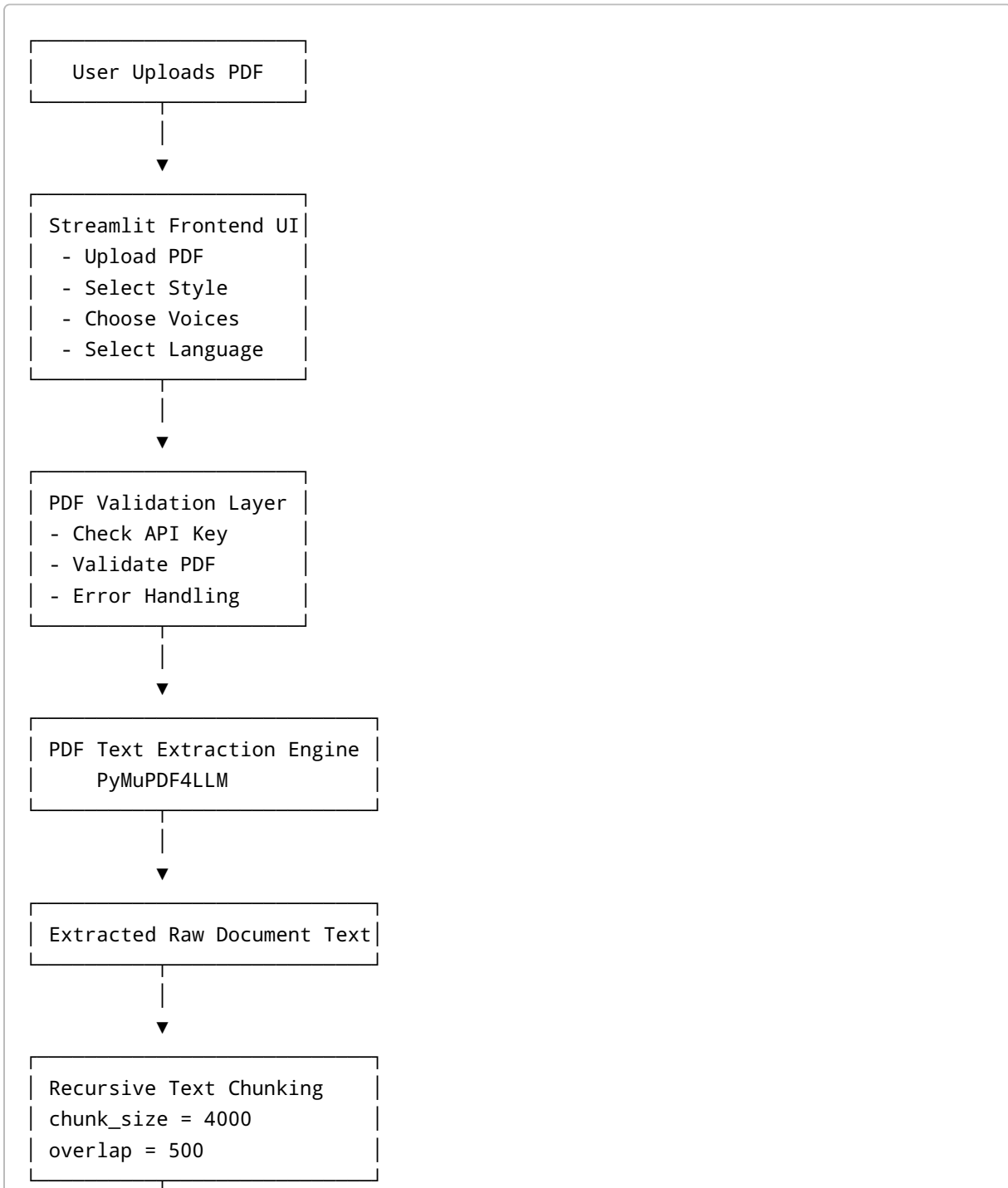
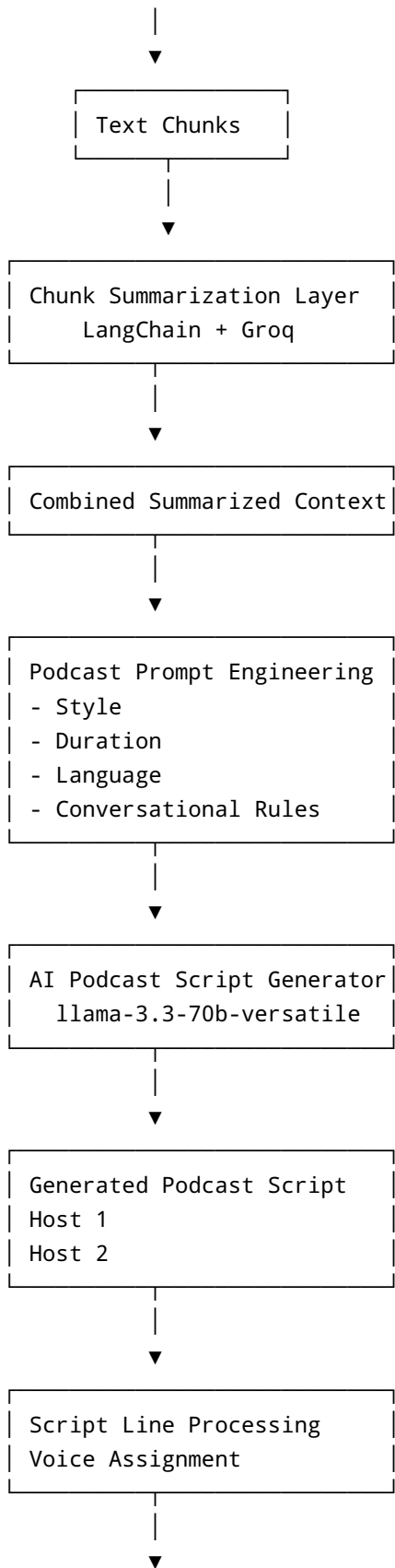
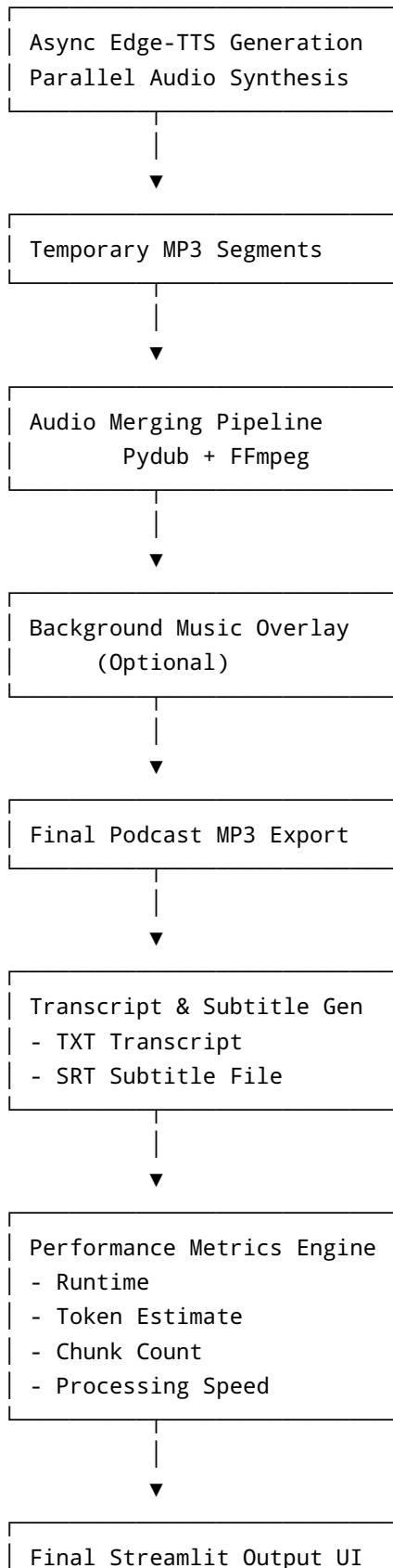


PDF to Podcast Generator — System Workflow

Complete System Architecture

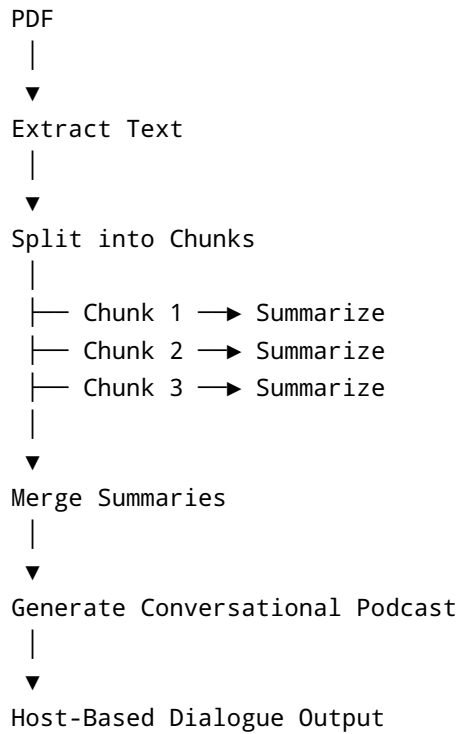




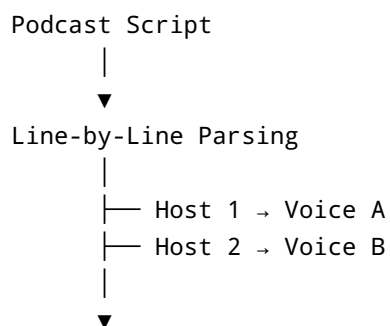


- Audio Player
- Downloads
- Metrics Dashboard

Detailed AI Processing Flow



Audio Generation Workflow



Edge-TTS Async Generation



Temporary Audio Files



Pydub Audio Merge



FFmpeg MP3 Export



Final Podcast Audio

Performance Optimization Workflow

Large PDF



Chunk Processing



Parallel TTS Generation



Reduced Runtime

Optimizations Used

- Async audio generation
- Cached model loading
- Cached PDF extraction
- Chunked LLM processing
- Parallel TTS synthesis



Metrics Collection Pipeline

PDF Extraction Time

+

Chunk Processing Time

+
LLM Generation Time
+
Audio Synthesis Time
+
Export Time
↓
Total Runtime

Error Handling Workflow

User Action
↓
▼
Validation Checks
|
|— Missing PDF
|— Invalid API Key
|— Empty PDF
|— FFmpeg Missing
|— Audio Export Failure
↓
▼
Graceful Error Messages

Multi-Language Support Pipeline

Selected Language
↓
▼
Prompt Customization
↓
▼
Language-Specific Script
↓
▼
Voice Model Selection
↓

▼
Localized Podcast Audio

Output Generation Flow

Generated Script
|
├─ MP3 Podcast
├─ TXT Transcript
├─ SRT Subtitles
|
▼
Downloadable Assets

Final System Overview

PDF + AI + TTS + Audio Processing
↓
Fully Generated Podcast